

Federated Machine Learning

Dataset Information

Cardiovascular Disease Dataset

- 70,000 instances
- 12 attributes

Binary Target Variable:

- Class 0 — No Cardiovascular Disease: 35,021
- Class 1 — Has Cardiovascular Disease: 34,979

Source:

<https://www.kaggle.com/datasets/salomemweluscherer/cardiovascular-disease-dataset>

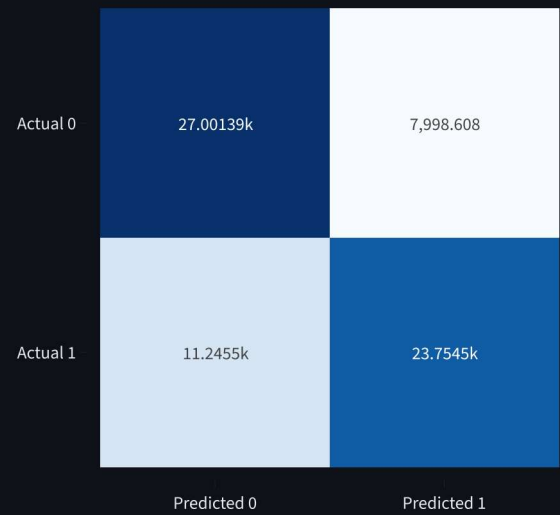
Run Federated Learning Pipeline

Pipeline Completed

Baseline Model Performance

	accuracy	precision	recall	f1	roc_auc
0	0.7253	0.7481	0.6787	0.7117	0.7252

Confusion Matrix (Baseline Model)



Hospital Performance

Hospital 1 Analysis

Decision: degraded_after_increment_retrained

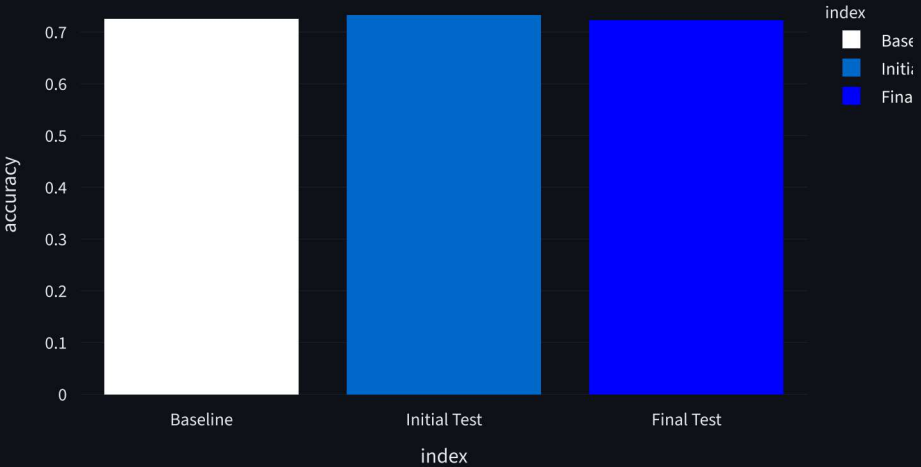
	accuracy	precision	recall	f1	roc_auc	samples
Baseline	0.7253	0.7481	0.6787	0.7117	0.7252	None
Initial Test	0.7327	0.7601	0.6912	0.724	0.7333	1500
Final Test	0.723	0.747	0.6738	0.7085	0.723	3000

Hospital 2 Analysis

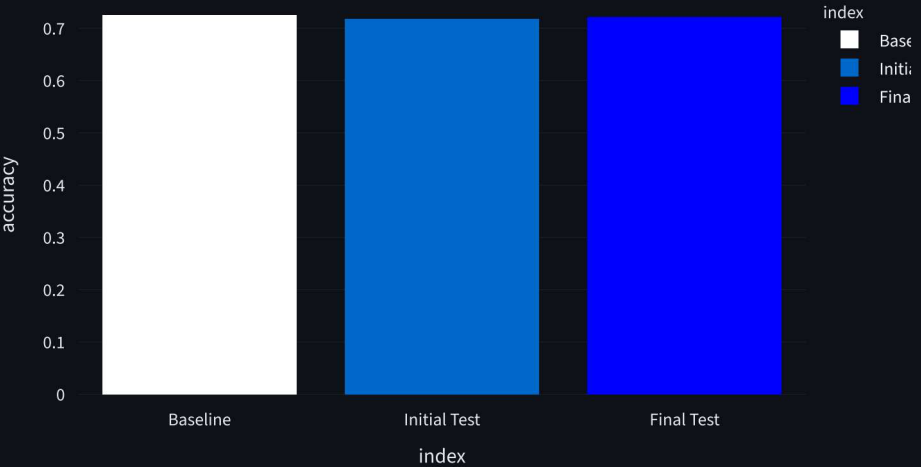
Decision: retrained_due_to_reduction

	accuracy	precision	recall	f1	roc_auc	samples
Baseline	0.7253	0.7481	0.6787	0.7117	0.7252	None
Initial Test	0.7177	0.7448	0.6662	0.7033	0.7179	3000
Final Test	0.7222	0.7494	0.6673	0.706	0.7222	4000

Hospital 1 Accuracy vs Baseline



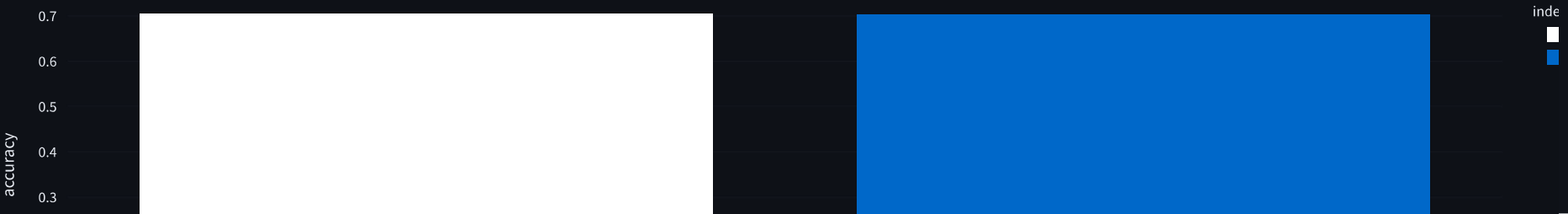
Hospital 2 Accuracy vs Baseline

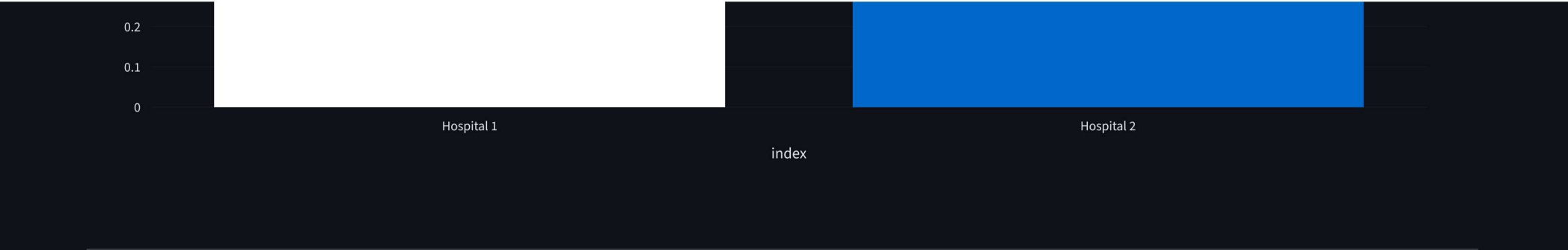


Global Model Performance (main_model_v2)

	accuracy	precision	recall	f1	roc_auc	samples
Hospital 1	0.705	0.7234	0.6631	0.692	0.705	3000
Hospital 2	0.703	0.7254	0.6528	0.6872	0.703	4000

Global Model Accuracy Across Hospitals

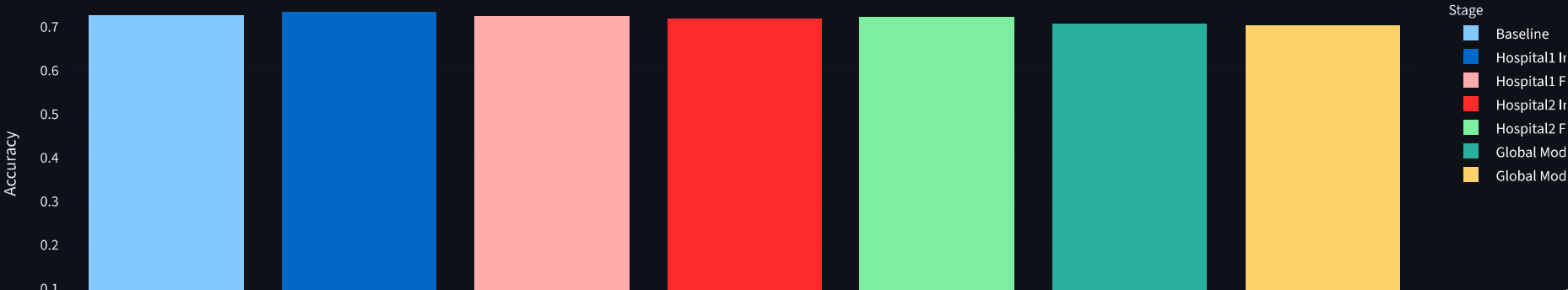


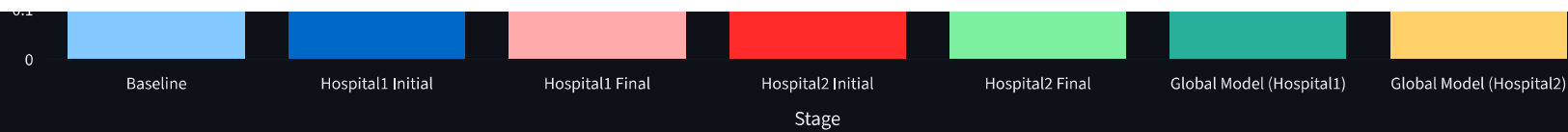


Accuracy Comparison

	Stage	Accuracy
0	Baseline	0.7253
1	Hospital1 Initial	0.7327
2	Hospital1 Final	0.723
3	Hospital2 Initial	0.7177
4	Hospital2 Final	0.7222
5	Global Model (Hospital1)	0.705
6	Global Model (Hospital2)	0.703

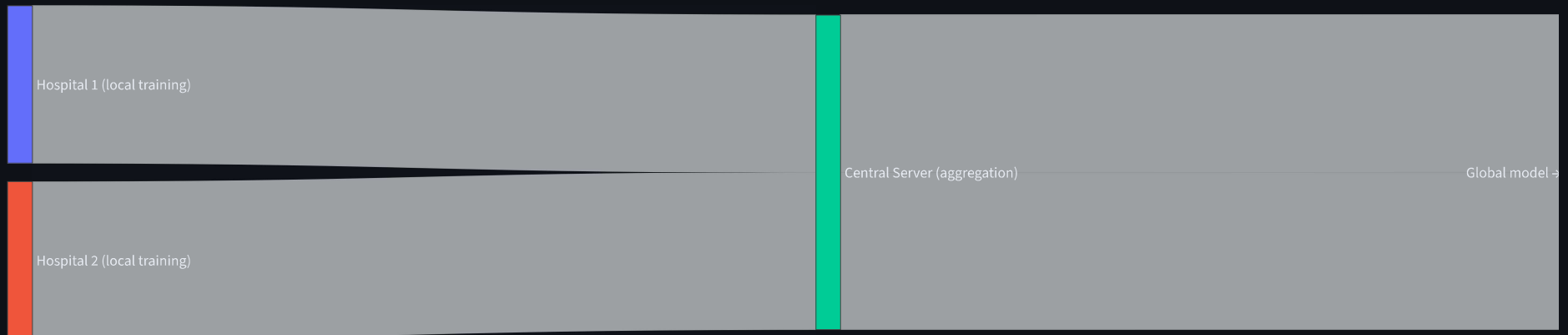
Accuracy Comparison Across Federated Learning Stages





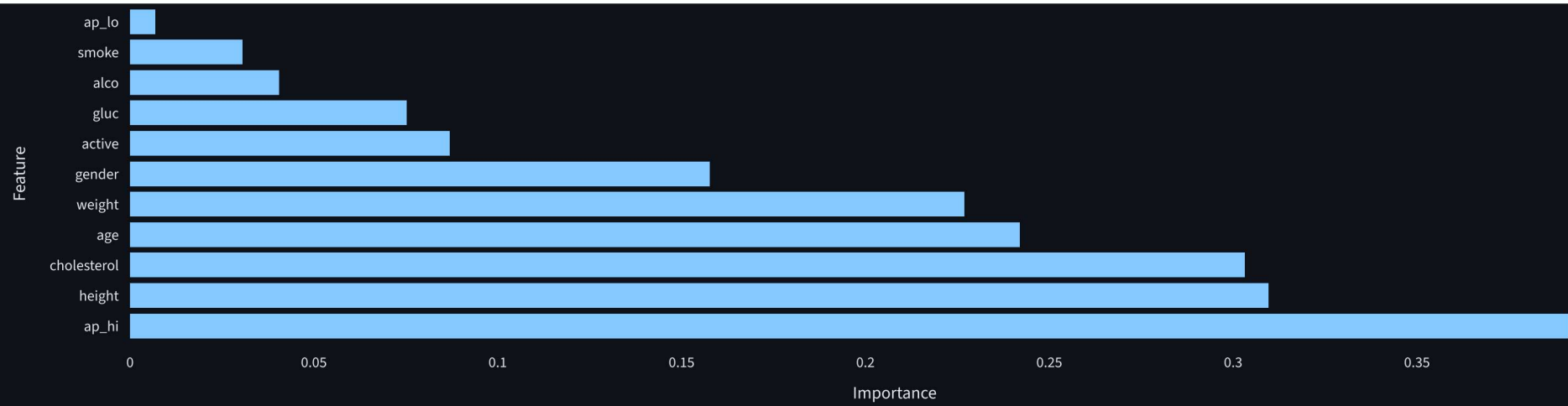
Federated Architecture (flow)

Federated Learning Data / Weights Flow (Sankey)



Explainability (SHAP)

Feature Importance for Cardiovascular Risk



	Feature	Importance
4	ap_hi	0.3951
2	height	0.3096
6	cholesterol	0.303
0	age	0.2419
3	weight	0.2269
1	gender	0.1576
10	active	0.0869
7	gluc	0.0751
9	alco	0.0405
8	smoke	0.0305